PROJECT DESIGN PHASE:

1. System Architecture Design

This defines how different components of the ServiceNow platform interact.

- **Front-End:** Service Portal (Widget-based UI for students to request books).
- **Back-End:** Custom Tables (`u_book`, `u_borrow_request`) and the Relational Database.
- **Logic Layer:** Flow Designer (The "brain" of the system that handles the approvals).
- **Integration Layer:** Email Notifications (SMTP) to keep users informed.

2. Database Schema (Data Model)

You must design the relationship between tables. For a library, a **One-to-Many** relationship is key: One user can have many borrow requests, but one specific physical book copy can only be tied to one active request.

Table Name	Field	Data Type	Reference / Note
Book	Title, ISBN, Author	String	Core inventory data.
Book	Availability	Choice	Available, Borrowed, Lost.
Borrow Request	Requester	Reference	Links to `sys_user` table.
Borrow Request	Book	Reference	Links to `u_book` table.

3. User Interface (UI) & User Experience (UX) Design

- **Service Portal Page:** A clean search bar to find books and a "Borrow" button.
- **Form Design:** Limiting the "Borrow Request" form to only essential fields (Book and Duration) to reduce friction.
- **Mobile View:** Ensuring the librarian can approve requests on the go via the ServiceNow Agent app.

4. Process Flow Design (The Logic Map)

This is a visual representation of the flow you were building in your screenshot.

1. **Submission:** User clicks "Submit."
2. **Validation:** Script checks: *Is the user's account active? Is the book on the shelf?*
3. **Approval Branch:** * Trigger "Ask for Approval" to Librarian.
    - **If Approved:** Update Book Status $\rightarrow$ "Borrowed"; Send "Ready for Pickup" Email.
    - **If Rejected:** Send "Request Denied" Email.
4. **Timer:** Wait 14 days $\rightarrow$ Send "Overdue" reminder.

5. Security & Access Design (ACLs)

Who can see what?

- **Read Access:** Everyone can see the list of books.
- **Write Access:** Only Librarians can edit book details or ISBNs.
- **Create Access:** Any authenticated student can create a "Borrow Request."

6. Error Handling Design

As seen in your initial screenshot error, a robust design must account for failures.

- **The "Error Handler":** If the Librarian doesn't respond within 48 hours, the request should auto-escalate or notify the Library Manager.
- **Mandatory Checks:** Ensure the system doesn't try to process a request that is missing a "Book Record" pill.

Design Artifacts to Create:

- **Entity-Relationship Diagram (ERD):** Showing how the tables connect.
- **Flowcharts:** To visualize the "If/Then" logic.
- **Wireframes:** Low-fidelity sketches of the Service Portal.